

□ 13 Descriptive statistics

Two characteristics of interest for a single variable are measures of central tendency and measures of spread.

A measure of central tendency determines an average or typical or representative value for the data.

A measure of spread determines how much variation or dispersion there is in the data.

- **Measures of central tendency**

Suppose days of *sun* data as an example for 8 farms is as shown.

5	3	6	8	4	9	2	3
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The mean (scale variables) (Colloquially called the average)

The mean is the total number of sunny days divided by the total number of farms. Thus, the mean = $40 / 8 = 5$

The median (scale or ordinal variables)

The median is the mid observation of the ordered (arranged in order from lowest to highest) data

In this example the ordered data is:

2	3	3	4	5	6	8	9
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To calculate the median, we take the mean of the 4th and 5th observation = $(4+5) / 2 = 4.5$.

(If we have an odd number of observations, we would simply take the unique middle value).

Note if the farm with the greater number of sunny days had 19 rather than 9 days, then the mean would change to 6.25 (Exercise) but the median would be unchanged. So, the median is not influenced by extreme values.

The mode (any type – i.e., scale, ordinal, nominal or binary variables)

The mode is the most commonly occurring observation. As more farms (2 in total) have 3 sunny days than any other number of days then the modal value is 3. The mode need not be, and frequently isn't unique.

- **Central tendency for categorical variables**

When measuring the central tendency of scale variable – either continuous or discrete, then the mean, the median and the mode can be used. What about categorical variables?

Example

Consider the grades achieved by leaving certificate students in mathematics in 2019. The data consists of thousands of observations – e.g., H4, H1, H7, H7, H2 etc. Clearly this is an ordinal variable (H1 is higher than H2 but we cannot say by how much).

It is plainly not possible to get the mean of such non-numerical data, but can we get calculate the median or the mode. To do so, we first we compile an ordered table of percentage frequencies and cumulative percentage frequencies as shown below.

	H1	H2	H3	H4	H5	H6	H7	H8
Percent freq.	6.0%	11.1%	17.0%	22.6%	21.8%	14.3%	5.4%	1.8%
Cum. Percent freq.	6.0%	17.1%	34.1%	56.7%	78.5%	92.8%	98.2%	100.0%

We can readily see that both the median is H4 as the middle value (i.e., 50%) is in this category. The mode is also H4 as the 22.6% of students attained this grade, more than more any other grade.

Example

Consider again the customer survey data and the variable capturing type of goods sold (tabulated above). This is clearly a nominal variable as there is no ordering electronics, jewellery and fragrances. And as the data cannot be ordered then we cannot find the middle or median value. But we can calculate the modal goods item sold to be jewellery as there are more items of this sold (42%) compared to either of the other two goods types.

- **Measures of spread**

Suppose the *rain* data for 8 farms is as shown.

31	16	25	24	17	29	22	NA
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The range (scale variables)

The range is $\text{max} - \text{min} = 31 - 16 = 15$

Quartiles and the interquartile range (scale or ordinal variables)

The first quartile Q1 for a scale variable is the value K of that variable such that 25% of observations are less than or equal to K. Similarly, the third quartile Q3 corresponds to the 75th percentile.

For example, in a sample of data recording the ages of retired persons we might calculate the first quartile to be 58.4 years and the third quartile to be 78.9 years.

Thus, 25% of these retirees have ages ≤ 58.4 and 75% have ages ≤ 78.9 .

Note that the middle 50% of ages are between 58.4 and 78.9 years – a range of 20.5.

This middle 50% range is called the inter-quartile range (IQR).

And half of this quantity is called the semi, inter-quartile range.

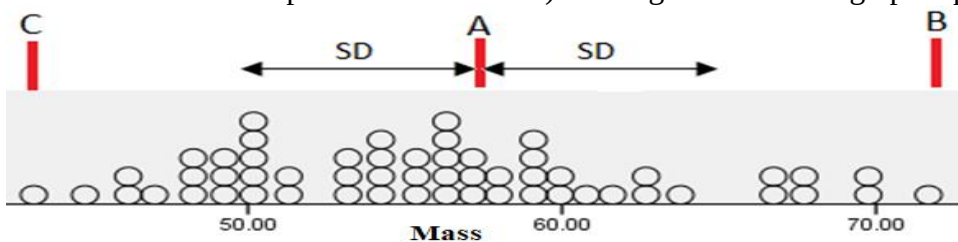
Quartiles (4), percentiles (100) and deciles (10) are all examples of quantiles. There are several different algorithms used (9 in R!) for calculating these.

The sample standard deviation s and the variance s^2 (scale variables)

The standard deviation (s or SD) is the primary way variation is measured in statistics. It measures the average deviation of a sample of data about the mean value. To get a sense of that this actually measures, consider the following fragment from a statistical report.

“The weights in a sample of animals were found to have a mean of 55.9 and a standard deviation of 6.81 kg.”
It is at once clear to most what the mean value conveys. But what about the SD?

To help address this, we retrieve the original data and construct (SPSS) a plot (called a dot plot – every individual datum is represented with a dot) and augment it with a graphic placed above the plot.



The average (mean) mass is about 56 kg, and some animals deviate little from this average (A)
I.e., the mass of some individuals is very close to the mean.

The maximum mass is about 72 kg, which is 16 kg above the mean,
And some animals deviate by as much as this from the average (B).

Similarly, the minimum mass is about 43 kg, which is about 13 kg below the mean.
And some animals deviate by as much as this from the average (C)

Note that when the data is not skewed, B and C will typically about equidistance from A.

So, some animals have minimum deviation (zero) from the average (A) – while others have maximum deviation (13 – 16 kg) from the average (B and C).

The standard deviation is a measure of *average* variation from the mean. Thus, a useful, if very approximate guide, is that the range is approximately 4 times the SD

Note that just as the mean and median will tend to be similar in the absence of extreme values or skewed data so too will the semi-interquartile range and the standard deviation.

So, after reading the report fragment that the mean and SD animal weights are about 56 kg and 7 kg respectively, we might mentally compute $56 \pm 2(7) = [42, 70]$ to be an interval that spans a sizable portion of the range of the data.

Example

Suppose we were given summary statistics for annual cost of home heating fuel to have a mean of €380 and a SD of €60, should we regard as usual, costs for two households of €490 and €240?

Answer

No and yes, (Exercise Why?)

Measures of spread in categorical variables

The standard deviation (or the range) may only be calculated for scale (i.e., numerical) data. But we can calculate the inter-quartile range (or semi-inter-quartile range) to get some measure of spread for ordinal data.

Example

Consider again the statistics for the grades achieved by leaving certificate students in mathematics in 2019.

	H1	H2	H3	H4	H5	H6	H7	H8
Percent	6%	11.1%	17.0%	22.6%	21.8%	14.3%	5.4%	1.8%
Cum. Percent	6%	17.1%	34.1%	56.7%	78.5%	92.8%	98.2%	100.0%

From the cumulative percent row, we can readily see that 25% of students obtained a H3 or better and 75% obtained a H5 or better – so the interquartile range is given by the range H5 to H3.

If we found the IQR for another subject to be the range H2 to H6 then we could say that there is more spread or variation in this other subject.

Note that in practice the IQR may often be of limited usefulness if there are few levels of an ordinal variable.

For example, a food/beverage portion might be classified as small, medium or large – thus it is an ordinal variable. Suppose that an analysis of the portions purchased by customers is described in the following percentage frequency table.

Portion	Small	Medium	Large
% frequency	30%	28%	42%

Determining (correctly!) that the inter-quartile range – or the middle 50% is given by “small” to “large” is not particularly informative.

We cannot find the IQR for a nominal variable as the data is not ordered. The only measure of spread we have for a nominal variable is the number of levels.

Example

If primary school A has pupils from 7 different countries, school B has pupils from 5 different countries and school C has pupils from 11 different countries then we could say that there is more spread / variation (we would probably call this ethnic diversity) in school A compared to B but less when compared to school C.

Example

A mobile phone operator who offers 10 different tariff plans would be showing more spread / variation (we would probably call this enhanced customer choice) than one that offers only 6 tariff plans.

□ Charts

Charts/plots are an effective way to convey information pertaining to data. There is a plethora of different chart types available – we will consider some guidelines as to how to make the selection.

Recall that the type variables may be classified as scale (continuous or discrete) or categorical (nominal, binary or ordinal). This top-level classification, scale or categorical (in most cases) as well as the number of variables being considered need to be considered when selecting a suitable statistical chart.

Our concern will be substantially with the business of selecting suitable charts and less with the mechanics of constructing them by hand – typically we use software to do so. The charts example below are all generated using one of Microsoft Excel, R, Minitab and SPSS (the last three are dedicated statistical software packages).

One of the first tasks that should be undertaken in any statistical investigation is to examine the data variable by variable, both to get a “feel” for the data – but also to identify potential errors. A suitable chart of each variable taken in turn is an effective way to achieve this.

Example

Suppose we collected the ages of 30 people employed in a firm but grouped them in a frequency table by the decade of their age, i.e., twenties, thirties etc. Such a table would involve two rows (or columns). One row would indicate an age range (e.g., 20 – 29) and the second would record the number of people (i.e., the frequency) with ages in that range (e.g., 6 in the 20 – 29 range etc.). This is an example of a frequency table of grouped data. A hypothetical example is given below, in raw form first and then grouped.

40	46	44	52	45	37	31	59	49	23
51	33	40	53	44	55	39	51	28	59
65	65	62	25	51	24	49	30	25	24

Range	20-29	30-39	40-49	50-59	60-69	Total
Frequency	6	5	8	8	3	30

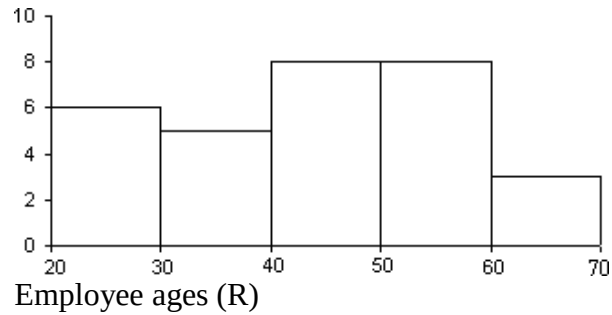
Take care that the ranges do not overlap. E.g., 20 – 30, 30 – 40.

We can get a much better idea of the spread of ages from this frequency table than we can from the raw data. This advantage is even more pronounced when we have a large data set.

The cost of course is that we have lost some information. For example, there are six people in the group who are in their twenties, but we cannot recover their exact age from the table.

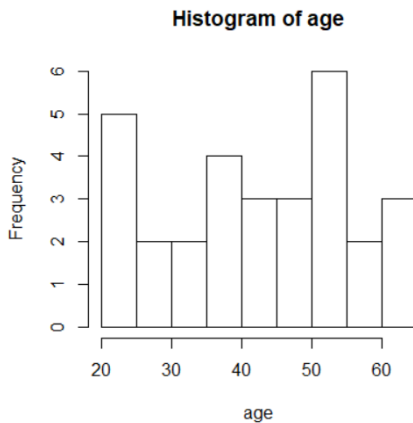
Histogram

We can represent our data graphically using a histogram (shown opposite). Each range in the frequency table is represented by a bar, the height of which is determined by the range of that frequency. As with the frequency table we can get immediately get a feel for the distribution of ages in this firm.

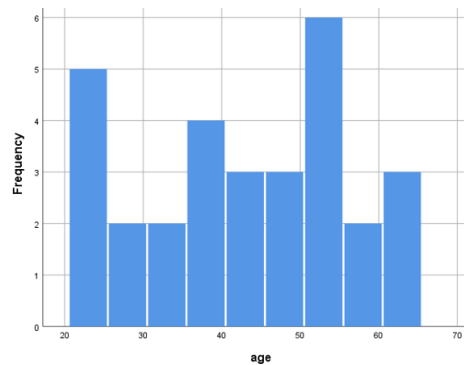


In the example above we say that there five bins and the bin width is 10. There are a number of algorithms used to decide, for a given data set, appropriate values for bin count and bin width, though they usually can be specified by the user. (In the example above, the bins were selected to have width 10).

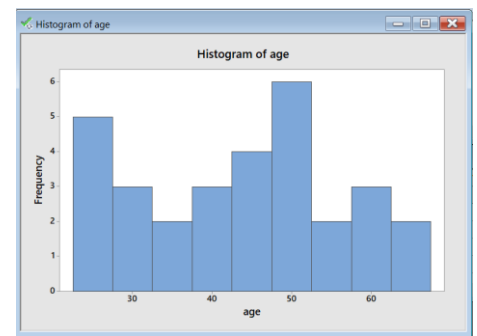
Different software might use different default algorithms leading to (slightly) different histogram plots. The default plots for the example above for R, Minitab and SPSS are shown below.



R



SPSS



Minitab

Stem and leaf plot

The advantage of frequency tables and histograms comes at a price, which is that there is some loss in information. For example, we can tell from the first histogram that 3 people have ages in the 60-69 age group but we don't know their specific ages. The stem and leaf plot offers the best of both worlds and is shown opposite.

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2 | 344558
3 | 01379
4 | 00445699
5 | 11123599
6 | 255
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Note that we have retained the detailed information on the ages of employees.

Employee ages (R)

But we also can see at a glance the spread of the ages by viewing the plot as a histogram rotated through 90°.

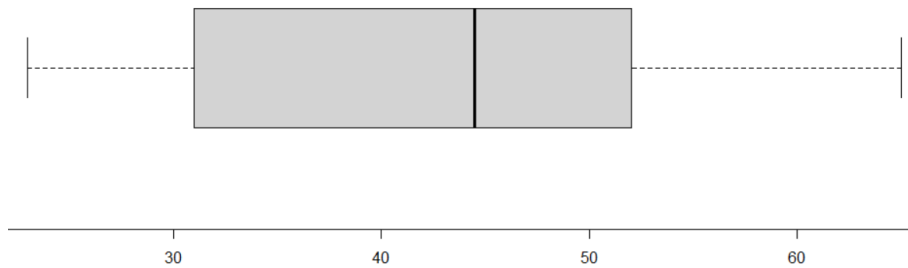
Boxplot

Earlier we considered how, for a scale variable we would find the first and third quartile Q1 and Q3, and earlier the median (Q2). If we take these three statistics along with the minimum and maximum observed value, we can compile a five-point summary for our scale variable, which we can arrange in order. For the age data we get:

Minimum	Q1	Median	Q3	Maximum
23	31	44.5	52.5	65

Note that 25% of ages are in the interval [23, 31), 25% in [31, 44.5), 25% in [44.5, 52.5) and 25% in [52.5, 65].

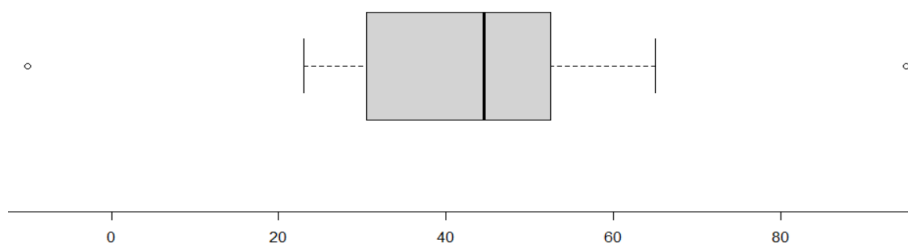
We can represent this five-point summary using a boxplot – such a plot (R) is shown below



The left of the box is at Q1, the right is at Q3. The solid line through the box is at the median. The lines out from the left and right of the box (the whiskers) extend to the minimum and maximum. The height of the box section is insignificant.

Most software that generates boxplots uses an algorithm to detect outliers (i.e., values that are either very large or very small relative to the rest of the data) and plots these values separately from the rest of the plot. Some distinguish further between what they call mild and extreme outliers.

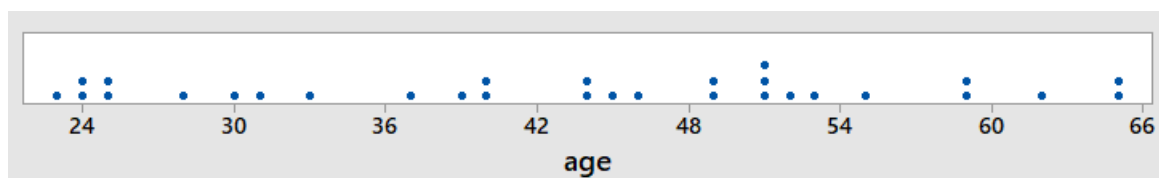
Suppose we augmented our employee age data with two additional value -10 and 95. The resultant boxplot shown in R would be as below.



Note that both of these additional values have been identified as outliers and plotted separately with ‘o’ symbols. It is important that outliers be identified in any analysis of the data. Often, they arise from errors (it is improbable that an employee might be aged 95 and impossible that they are aged -10). Even if they are genuine values, it is still important that they be identified (they may be distorting our findings)

Dot plot

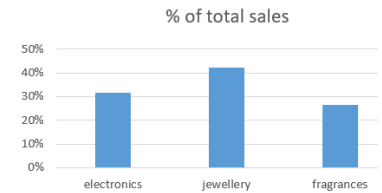
A dot plot is a plot of dots on a scale axis where each dot represents one datum (or sometimes more). An example was given above when standard deviation was being described. The dot plot may be preferred to a histogram or boxplot when the data set is small. A dot plot of *age* (Minitab) is shown below.



For qualitative or categorical data there are two popular charts used to represent the data.

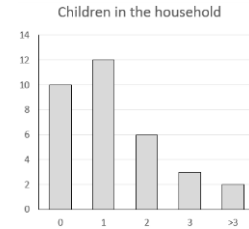
Bar charts

We have already seen a bar chart for goods type in the customer survey data. This is reproduced opposite (Excel). The chart is superficially similar to a histogram but with a bar chart we leave a space between adjacent bars.



Bar charts may also be used for discrete scale data, especially if there a small number of distinct values.

For example, the bar chart opposite (Excel) showing, for a household survey, the number of children in the household, represents this discrete data quite satisfactorily.

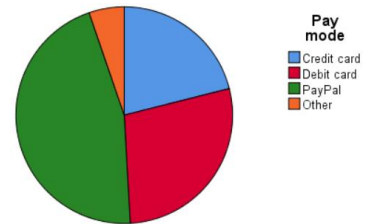


This is an example of a frequency bar chart – the previous example was of a percentage frequency bar chart.

Pie charts

A pie chart of the payment mode for the customer survey data (SPSS) is shown.

A pie chart allows us to more readily see the proportion that one mode contributes to the total (e.g., we see almost half pay using PayPal)



But is it less easy to compare relative proportions especially if the size of each slice as a percentage is not shown on the chart (the default in SPSS). For example, we can (just about!) determine that there are more debit card users than credit card users – but this would be much easier to see on a bar chart.